SVKM's NMIMS · Mukesh Patel School of Technology Management & Engineering, Mumbai

Modern Day Robotics & Its Industrial Applications (MDRIIA - 702CO0E012) · B.Tech Computer Science & Business Systems (CSBS), in partnership with TCS · AY 2026–27 Odd Semester

GROUP 11 TECHNICAL DOSSIER

Capstone ICA (20 marks): CDIO Robotics Engineering & MuJoCo Physics 10 · CSBS Techno-Economic Business Systems Analysis 5 · Studio Showcase & Viva Voce 5

Group · project	Group 11 · Autonomous Solar PV Inspection AMR
Monorepo folder	https://github.com/sunny-nanade/NMIMS-MPSTME-MDRIIA-2026/tree/main/Group_11_Daneeka
Submit (3 places)	(1) PDF on MS Teams, turned in by EVERY member · (2) this filled Word file AND its PDF committed to Group_11_Daneeka/docs/MDRIIA_Group_11_Technical_Dossier.docx / .pdf · (3) LinkedIn video URL in Section 5. Due Monday, 26-Oct-2026, 23:59 IST. Studio Showcase, Expo & Viva Voce: Saturday, 31-Oct-2026.
No slides	There is no PPT for MDRIIA. At the Expo and in your video you present the live MuJoCo simulation, your business sensitivity graphs, and a webcam picture-in-picture of every member.
How to fill this file	Use Microsoft Word (desktop). Choose answers from the 'Select ▾' drop-downs and type into the 'type here' fields. Do not delete or retype them and do not edit this file in Google Docs: the fields are how your answers are read.

Student name	Roll No.	SAP ID	Capstone role(s)	Specialisation (finalised titles sheet / monorepo)
Daneeka Abhijeet Roy	E057	70362400081	Lead Robotics Systems Architect + Autonomous Navigation / Vision Specialist + CSBS Business & Techno-economic Systems Analyst + Telemetry QA Lead	Lead Autonomous Systems Architect & Sensor Anomaly Modeler

Roll numbers and SAP IDs are from the official student list. Tell Dr. Nanade by Tuesday, 20-Oct-2026 if any name, ID or role is wrong. Groups with fewer than four members combine roles as shown.

SECTION 1 · Problem Bottleneck, Industry Standards & Aalborg Research Title *Component 1 · CDIO Robotics (10)*

Authorised Aalborg title	How can autonomous mobile-robot navigation and sensor-based anomaly detection improve the speed and accuracy of locating defective solar panels in large-scale solar installations compared with manual inspection?
Original draft topic	(new group: split in October 2026)
Why the title changed	Group created when the original team was split; title taken from the group's monorepo README.

1.1 Baseline friction: claim verification table

Each statement was carried over from your titles sheet or your monorepo scripts. Before you use it anywhere (dossier, video, paper), find the primary source yourself and record it, or correct or delete the claim. Statistics: cite the original study, not a blog or an AI answer.

Claim (as carried over)	Faculty pre-audit note	Primary source you verified (author, year, venue, DOI/URL)	Source type	Keep / Correct / Delete
Manual inspection takes 8.5 h per MW versus 1.6 h per MW for the AMR.	*Hard-coded in solar_pv_roi.py with no source. Find an O&M source for the manual figure; the AMR figure must come from your simulation.*	Solar Builder magazine, "This Drone for Hire: Meeting the Challenges of Solar O&M with Aerial Thermography" (G. Shambo, Borrego O&M, quoting an unnamed study): manual 10 h/MW. Secondary trade press, read 30-Oct-2026; not peer-reviewed.	Other web page	Correct
Cost-parity ratio κ = 0.176, labour reallocation 78.5%, payback 13.4 months.	*These are typed-in results, not computed. Replace with your own calculation.*	No source: these were typed-in results. Replaced by values computed in analytics/solar_pv_roi.py from our own Monte Carlo runs (kappa 0.463, reallocation 83.7 %, payback 67.1 months; see 3.1).	No source found	Delete
A hotspot is a module with ΔT ≥ 15 °C.	*Check IEC TS 62446-3 (outdoor infrared thermography of PV) for its temperature-difference classes and cite the clause.*	No primary source found. IEC TS 62446-3 was not read in full (clause/table not verified). 15 K is used only as the BASELINE rule; the proposed detector uses an irradiance-normalised relative excess instead.	No source found	Correct

1.2 Governing industry standards

Decide for each standard whether it governs your system, and name the clause or requirement you design against (cite number, part and year). 'No' is a correct answer when you can say why. The hint is the faculty view; you may disagree with a reason.

Standard	Scope	Faculty hint for your project	Applies?	Clause / requirement you use, or why not
ISO 10218-1/-2	Safety requirements for industrial robots (robot, and robot integration)	*No*	No	Covers industrial manipulators and cells; our AMR is a mobile inspection robot with no manipulator.
ISO 3691-4	Driverless industrial trucks / AMRs and their systems: safety requirements	*Partly: outdoor AMR safety concepts*	Partly	Safety concepts only (stop on obstacle, speed limit, no personnel contact). Front range stop is simulated; no certification claimed. Clause numbers not verified.
ANSI/RIA R15.06	US national adoption of ISO 10218 (industrial robot safety)	*No*	No	US adoption of ISO 10218 (industrial robots); same reason as above.
IEC 60601-1	Medical electrical equipment: basic safety and essential performance	*No*	No	Medical electrical equipment; not relevant.
Domain-specific	IEC TS 62446-3 (infrared thermography of PV modules and plants)	*Recommended: check it*	Yes	IEC TS 62446-3 guides outdoor IR thermography of PV (irradiance gate, viewing geometry, emissivity). Our 600 W/m2 gate follows its recommendation but the exact clause/table was not read and is unverified.

1.3 Documented baseline friction (who struggles, with what, how it is measured today)

Baseline metric and its value today (with unit)	Manual PV inspection effort 10 h/MW per survey (technician hours).
Source of the baseline value (author, year, DOI/URL)	Solar Builder magazine, G. Shambo (Borrego O&M), "This Drone for Hire" (secondary, trade press, unnamed underlying study); read 30-Oct-2026. Most uncertain input: varied +/-30 % in 3.1.
Who carries the cost today (role / stakeholder)	O&M technicians / site O&M manager (labour cost of manual inspection).

SECTION 2 · MuJoCo Rigid-Body Physics Modelling & Python Closed-Loop Control *Component 1 · CDIO Robotics (10)*

MuJoCo mechanism (titles sheet)	Differential-drive inspection AMR with pan-tilt thermal/optical sensor head travelling along PV array rows
MJCF model	Group_11_Daneeka/models/solar_defect_amr.xml
Controller	Group_11_Daneeka/src/solar_defect_detector.py
Contact & solver settings in the model today	timestep 0.002 s (500 Hz) · wheel friction 1.2 / 0.05 / 0.002 · panel 0.4 / 0.005 / 0.0001 · solref 0.005 1.0 · solimp 0.9 0.95 0.001 0.5 2 · 5 actuators (2 wheels, pan, tilt, +1)
Core state machine (FSM)	ROW_FOLLOW → SCAN (pan-tilt sweep) → ANOMALY_FLAG (ΔT above threshold) → GEOTAG → NEXT_ROW → RETURN_TO_DOCK
Equations of motion & control	Newton-Euler (rigid body): $m\cdot\dot{v} = \Sigma F$, $I\cdot\dot{\omega} + \omega \times (I\cdot\omega) = \Sigma\tau$; multibody form $M(q)\cdot\ddot{q} + C(q,\dot{q})\cdot\dot{q} + g(q) = \tau + J^T\cdot f_{ext}$ Differential drive: $\omega_{R,L} = (v \pm \omega\cdot b/2) / r$; row-following PID on lateral offset e_y Hotspot flag if $\Delta T = T_{cell} - T_{mean,row} \geq \Delta T_{th}$; localisation error = $\|p_{flagged} - p_{true}\|$ PID tracking: $u(t) = K_p\cdot e(t) + K_i\cdot\int e\,dt + K_d\cdot\dot{e}(t)$ Damped least-squares IK (manipulators): $\Delta q = J^T\cdot(J\cdot J^T + \lambda^2\cdot I)^{-1}\cdot\Delta x$

Faculty code audit: src/solar_defect_detector.py is still a TODO stub that only prints a message: implement navigation, detection and CSV logging.

Faculty code audit: analytics/solar_pv_roi.py hard-codes its outputs (kappa = 0.176, payback 13.4 months). Replace them with formulas and sourced inputs.

2.1 Model and controller settings you actually used (from your final code)

MJCF timestep (s) in the final model	0.002 s (500 Hz)
Main contact friction (sliding / torsional / rolling)	Wheel-floor 1.2 / 0.05 / 0.002 (sliding / torsional / rolling); casters 0.01 with priority=1
solref (timeconst dampratio) and why you chose it	solref 0.005 1.0 (stiff, critically damped); solimp 0.9 0.95 0.001 0.5 2. Stiff enough that wheels do not sink, stable at 2 ms.
Controller law used	PID
Final gains (e.g. Kp, Ki, Kd or λ) with units	Lane PID Kp 1.4, Ki 0.15, Kd 0.25 on cross-track [m -> rad/s]; heading k_psi 2.2 [1/s]; wheel PI Kp 2.5, Ki 12 [1/s], feed-forward 0.05
FSM states implemented in code (list)	INIT, ROW_FOLLOW, SCAN, ANOMALY_FLAG, GEOTAG, NEXT_ROW, RETURN_TO_DOCK, DONE
Model loads and runs: python -c "import mujoco; mujoco.MjModel.from_xml_path(...)"	Yes

2.2 Authorship of each script

Several scripts were provided as starter scaffolds. Mark honestly what you wrote, changed, or used as given. Viva questions target the code you claim.

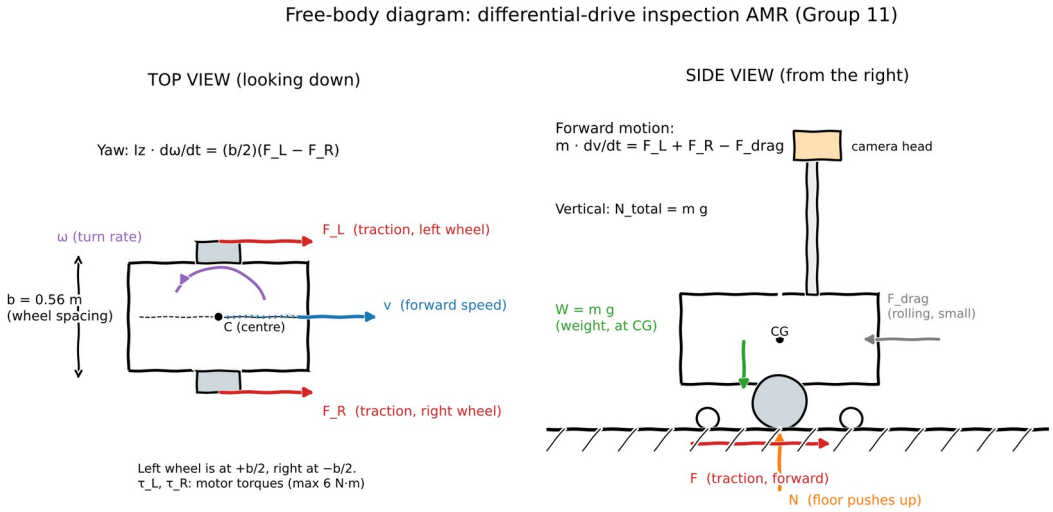
File	Written / Modified / Used as given	Main author (Roll)	What you changed or added (1 line)
models/solar_defect_amr.xml	Written by us	E057	Single-source MJCF generated by models/generate_mjcf.py; fixed caster friction (priority=1) and slab alpha/thickness. AI-assisted (Claude); run and reviewed by E057.
src/solar_defect_detector.py	Written by us	E057	InspectionMission FSM, sensing, navigation and detector integration (src/*). AI-assisted (Claude); run and reviewed by E057.
analytics/solar_pv_roi.py	Written by us	E057	Replaced hard-coded 8.5/1.6/0.176/78.5/13.4 with computed NCRI, kappa, payback, M/M/1 and M/M/c, +/-30 % sensitivity. AI-assisted.
analytics/run_evaluation.py	Written by us	E057	Held-out Monte Carlo (30 seeds x 2 conditions) to the real-trials CSV.
src/defect_detection.py	Written by us	E057	Baseline vs proposed detector; fault injection kept separate (test-enforced).

2.3 Key code snippet (≤ 25 lines) implementing your FSM or control law:

```
src/navigation.py, LaneFollower (lines 107-125): cross-track PID plus heading law, the row-following
controller
 107  class LaneFollower:
 108      """Cross-track + heading law about a straight lane. direction=+1 east, -1 west."""
 109
 110      def __init__(self, k_psi=2.2, y_pid: PID | None = None):
 111          self.k_psi = k_psi
 112          self.y_pid = y_pid or PID(1.4, 0.15, 0.25, out_limit=1.0, i_limit=0.4)
 113
 114      def reset(self):
 115          self.y_pid.reset()
 116
 117      def command(self, est: PoseEstimator, y_lane: float, direction: int, v_des: float, dt:
float):
 118          th_ref = 0.0 if direction > 0 else math.pi
 119          e_psi = wrap(est.theta - th_ref)
 120          e_y = est.y - y_lane
 121          corr = self.y_pid.step(e_y * direction, dt)          # + => turn right when heading east
 122          w = -self.k_psi * e_psi - corr
 123          w = max(-rp.OMEGA_MAX, min(rp.OMEGA_MAX, w))
 124          v = v_des * max(0.2, 1.0 - 1.5 * abs(e_psi) - 0.6 * abs(e_y))
 125          return v, w
```

2.4 Free-body diagram and one equation worked with your own numbers and units:

FBD: chassis with two driven wheels (radius r, track b), caster skids, contact forces F_L, F_R at the wheels, weight m*g.
Differential drive: v = r(w_R + w_L)/2, w = r(w_R - w_L)/b; wheel torque tau = J_w*dw/dt + r*F_x.
Lane law: w_cmd = -k_psi*e_psi - PID(e_y), k_psi = 2.2 1/s, PID(1.4, 0.15, 0.25); e_y = y - y_lane, e_psi = theta - theta_ref.
Worked example: e_y = 0.05 m, e_psi = 0.02 rad, integral and derivative ~ 0 -> w = -2.2*0.02 - 1.4*0.05 = -0.114 rad/s (turn toward the lane). With b = 0.56 m, r = 0.12 m (robot_params.py): w_R - w_L = w*b/r = -0.53 rad/s.
In telemetry: columns lane_y_ref, cross_track_err, heading_err (results/telemetry/demo_proposed_seed3009_500Hz.csv). FBD figure below.



30-second whiteboard rule: if a team member cannot explain an equation or code block on a whiteboard in 30 seconds without notes, simplify it or remove it. It will be asked in the viva.

SECTION 3 · CSBS Techno-Economic Business Systems Modelling *Component 2 · CSBS Techno-Economic (5)*

Business script	Group_11_Daneeka/analytics/solar_pv_roi.py (rename or add *_roi.py if you restructure it)
CSBS business case (titles sheet)	Inspection hours per MW (manual vs AMR), labour reallocation, dimensionless cost-parity ratio and payback for a utility-scale PV plant
Constants hard-coded in the script today	manual_inspection_hours_per_mw = 8.5, amr_inspection_hours_per_mw = 1.6, kappa = 0.176, labor_reallocation_pct = 78.5, payback_months = 13.4 → each one needs a source in 3.1 or must be computed from your simulation.
Owner	CSBS Business & Techno-economic Systems Analyst: E057. Tags in this section carry that roll number.

Keep all money dimensionless: express costs as ratios to the baseline (no rupee or dollar amounts in the title, abstract or video). Every input needs a source or must be a clearly labelled assumption that you vary in the sensitivity graph.

3.1 Business model: formulas, your inputs and your results

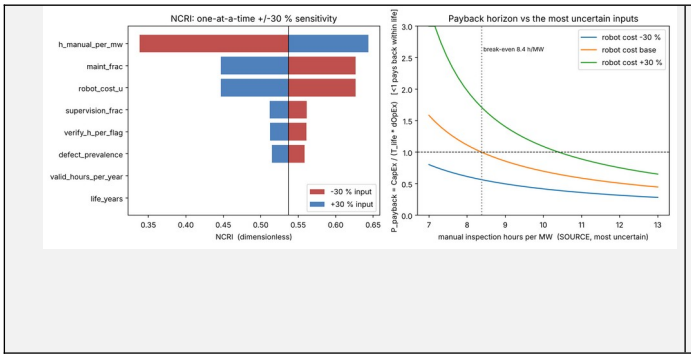
Quantity	Formula (define every symbol in your script)	Inputs used, with source or 'assumption'	Result (with unit)
Labour reallocation H_reclaimed	H = technician inspection hours per MW reallocated (technician-hours/MW)	h_manual 10 h/MW (secondary source); robot-attributed technician hours from the Monte Carlo runs (supervision + verification of flagged modules); 125,000 modules = 50 MW; all else labelled ASSUMPTION in the Inputs dataclass	8.37 h/MW reallocated (83.7 % of 10 h/MW)
Payback horizon P_payback (dimensionless)	P_payback = CapEx / (T_life · ΔOpEx_annual); P < 1 ⇒ pays back within life T_life. Also report T_payback = CapEx / ΔOpEx in months.	CapEx 3,000 u (2 robots, assumption); dOpEx 536.7 u/yr; T_life 8 years (assumption)	P_payback = 0.699 (<1: pays back within life); 67.1 months
Normalised Cost Reduction Index NCRI	NCRI = (C_baseline − C_robot) / C_baseline (0–1); cost-parity ratio κ = C_robot / C_baseline = 1 − NCRI	C_baseline 1,000 u/yr (manual); C_robot 463.3 u/yr (robot hours 548.3, supervision 82.3, verification of 1,351 flagged modules 81.1, maintenance 300)	NCRI = 0.537 (running cost); kappa = 0.463. Including the robot purchase cost spread over 8 years, total yearly cost falls about 16 %. Baseline controller: NCRI = -1.51 (false alarms make it costlier than manual)
Queuing delay	M/M/1: flagged modules (λ per day) awaiting technician repair (μ per day). ρ = λ / (c·μ) < 1; M/M/1: W_q = ρ / (μ − λ); M/M/c: W_q = C(c, λ/μ) / (c·μ − λ) (Erlang C)	lambda = 6.66 flagged modules/day (from sim flag rate); mu = 8 repairs/tech/day (assumption); c = 2	M/M/1 rho 0.833, Wq 0.623 d; M/M/2 rho 0.416, Wq 0.026 d
Sensitivity analysis	Vary the most uncertain input ±30 % (or across your scenario range) and plot P_payback and NCRI; report the break-even value.	Each input varied +/-30 %: h_manual, robot cost, defect prevalence, supervision fraction etc. (figure below)	Most sensitive: h_manual. At -30 % (7 h/MW) P_payback = 1.58 (does NOT pay back); break-even h_manual = 8.38 h/MW (P=1)

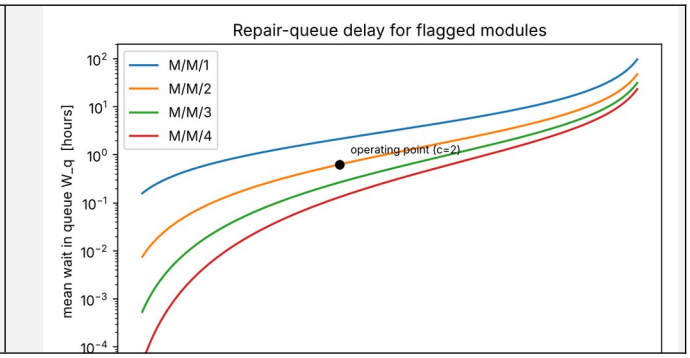
Queuing model used	M/M/c
λ, μ, c values used (with units)	lambda 6.66 /day, mu 8 /tech/day, c = 2 (M/M/1 shown for comparison, rho 0.83)
Payback in months (T_payback)	67.1 months (all money dimensionless "u")
Script output reproduces every number above when re-run	Yes

3.2 FMEA: failure modes, effects and risk priority (RPN = S × O × D, each 1–10)

Two failure modes are seeded from your project; add at least two more. Score before and after your mitigation (design change, control limit, sensor, procedure). The post-mitigation RPN must be justified by something you can show in the simulation or the code.

Failure mode	Effect on mission / business	S	O	D	RPN pre	Mitigation (what you changed)	RPN post	Shown in
Missed hotspot because of sun glare / sensor angle	Hotspot missed or false alarm; wrong repair dispatched	8	10	8	640	Glint mask + lift selection, 3-frame median, emissivity compensation	128 (O 4, D 4)	Both
AMR drifts into inter-row gap or kerb	Mis-registered scans, possible collision, mission abort	9	10	7	630	Range-pair + fiducial fusion, lane PID, front-range stop	108 (O 4, D 3)	Both
Wrong module identified / geotag on wrong module	Wrong module repaired; lost technician time	7	10	8	560	Fiducial position fixes, per-row association check	196 (O 7, D 4)	Both
Thermal assessment below valid irradiance (low sun / cloud)	Assessment made outside valid conditions, false results	6	10	7	420	Irradiance gate, wait/retry then mission deferral	48 (O 4, D 2)	Both
Thermal sensor noise, spikes, frame drop-outs	Spurious flags / missed hotspot from noisy frames	6	10	7	420	3-frame temporal median (residual high: spikes are not fully rejected)	180 (O 10, D 3)	Both





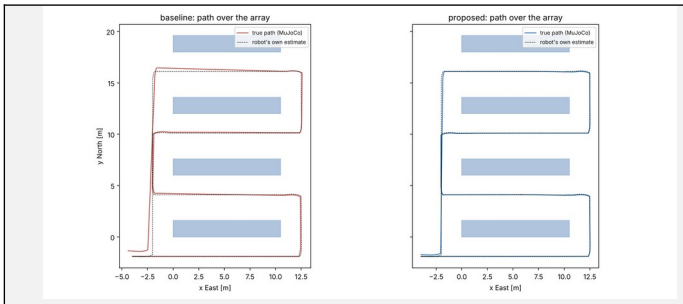
SECTION 4 · Simulation Telemetry, Monte Carlo Testing & Empirical Validation *Component 1 · CDIO Robotics (10)*

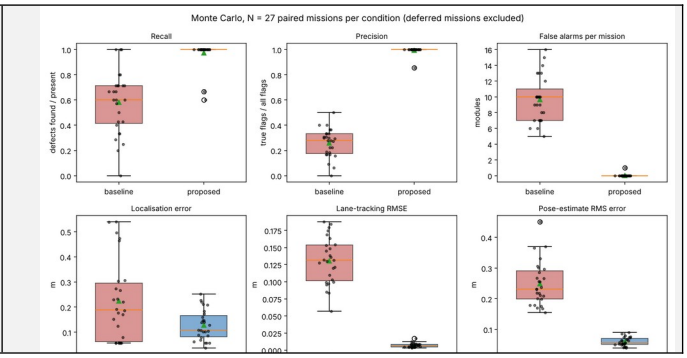
Evaluation protocol (titles sheet)	Run N ≥ 30 randomised inspection missions over a simulated array with seeded defective modules. Log time per MW, detection precision / recall, localisation error (m) and path-tracking RMSE.
Telemetry standard	Log at 500 Hz (MJCF timestep 0.002 s, or every 2nd step at 0.001 s): time, state (q, q̇ or pose, velocity), reference, tracking error, actuator torques/forces, FSM state, events. One CSV per run or one CSV with a run_id column.
Monte Carlo	N ≥ 30 runs per condition (more if your protocol says so). Randomise disturbances and initial conditions from stated distributions; record every random seed.
Required measures	Tracking RMSE = $\sqrt{(\Sigma\|p - p_{ref}\|^2 / n)}$ on the planned row-following path vs MuJoCo AMR position; control effort E = $\int \tau^2$ dt (sum of $\tau^2 \cdot \Delta t$ over all actuators); your primary task metric from the protocol.
Statistics	Compare the proposed controller with a baseline condition: Student's / Welch's t-test (two-sided, α = 0.05), Cohen's d = $(\bar{x}_1 - \bar{x}_2)$ / s_pooled, and a 95 % confidence interval.
Where real trial data goes	Group_11_Daneeka/analytics/*_real_trials.csv, produced by your controller. Files named SIMULATED_* are starter targets, never results.
Owner	Telemetry QA Lead: E057.

Baseline condition you compare against	Odometry + gyro only, 1 thermal frame, fixed 15 K DeltaT rule, no irradiance gate
Monte Carlo runs per condition (N)	30 per condition (seeds 3000-3029, held out; tuning used 1000-1029). 3 proposed runs deferred for low irradiance; paired stats use N = 27 (26 for localisation / type accuracy).
Randomised parameters and their distributions	Weather (clear/hazy/partly cloudy), start time U(10,14) h, defect count/kind/cell/severity, confounders, wheel calibration, gyro bias, sensor noise (seeded)
Logging rate actually achieved (Hz)	500 Hz (one row per 0.002 s step); full CSV kept for demo runs, per-mission summaries for all 60 runs
Random seeds recorded for every run	Yes
Real-trials CSV path(s) in the repo	Group_11_Daneeka/analytics/solar_inspection_real_trials.csv (also scenario_real_trials.csv, tuning_trials_seeds1000.csv)

Measure	N	Baseline mean ± SD	Proposed mean ± SD	Unit	Better? (direction)
Tracking RMSE	27	0.1305 +/- 0.0341	0.0071 +/- 0.0030	m	Lower is better
Control effort $\int \tau^2$ dt	27	889.2 +/- 6.2	887.1 +/- 7.7	N^2 m^2 s (no significant difference, p = 0.26)	Lower is better
Recall (defects found / present)	27	0.582 +/- 0.250	0.973 +/- 0.098	fraction	Higher is better

Test used	Welch's t-test
t statistic, degrees of freedom, p-value (primary metric)	t = 7.57, df = 33.9 (Welch), p = 8.7e-9
Cohen's d (primary metric)	d = 2.06
95 % confidence interval of the difference	+0.391 (95 % CI 0.286 to 0.496), proposed minus baseline





DATA PROVENANCE DECLARATION (mandatory). Reporting generated target numbers as experimental results is research misconduct and scores zero for this component.

Every number in Section 4 comes from MuJoCo runs of OUR controller, logged by our code	Yes
No value comes from SIMULATED_* files, generate_paper_figures.py or any np.random 'target' data	Yes
Our plots re-generate from the *_real_trials.csv files in the repo	Yes
Every input in Section 3 has a source or is labelled as an assumption	Yes

SECTION 5 · LinkedIn Video Showcase, AI Verification Audit & Peer Review *Component 3 · Showcase & Viva (5)*

LinkedIn post URL	type here
Video length (mm:ss)	type here
Video shows the live MuJoCo simulation (not slides)	Yes
Video shows our business sensitivity graph	Yes
Webcam picture-in-picture: every member appears and speaks	Yes

5.1 AI Interaction Audit Log: three cases where AI gave wrong robotics or economic data

Examples: a wrong MuJoCo attribute or API call, an invented statistic, a correct paper given the wrong journal, an economic formula with the wrong units. Show how you checked each one with a textbook, the official documentation or the original paper.

#	AI tool	What we asked & what AI said	How we caught it	What was actually true (with source)	Error type	Verdict
AI-01	Claude	Asked for a working robot model; it assumed caster friction 0.01. Robot would not move (torque 6 Nm, wheel speed 0.04 rad/s).	Ran / tested the code or simulation	MuJoCo combines two geoms' friction with max, so floor 1.2 won; caster tangential/normal force ratio was 0.33. Fix: priority="1" (MuJoCo docs, contact).	Wrong code / MuJoCo API	Corrected
AI-02	Claude	Said range sensing worked; mj_ray returned nothing on the module slab (alpha 0 skipped) so the robot silently dead-reckoned and drifted 18 cm by row 4.	Ran / tested the code or simulation	Per-row cross-track error grew 0.7 -> 18 cm; a ray test with all groups hit, group 3 alone did not. Fix: alpha 0.004 and 0.5 mm slab thickness (removed a 5.9 cm offset).	Wrong code / MuJoCo API	Corrected
AI-03	Claude	Starter files carried 8.5 h/MW, 1.6 h/MW, kappa 0.176, 78.5 %, 13.4 months with no source, plus unverified DOIs.	Checked the original source	No primary source; replaced by computed values and a sourced (secondary) 10 h/MW. Also: fiducial fixes never fired (row-1 bug) until a stress test showed poor recall; all runs repeated.	Wrong number or statistic	Rejected

5.2 Peer review: watch both assigned LinkedIn videos and score them (0–2 per criterion)

Group reviewed	Problem	Physics	Business	Data	Clarity	Best moment (mm:ss)	One question for the viva
Group 12: Autonomous Pipeline Surveillance UAV	2	2	2	2	2	2:25	How did you measure defect localisation latency, and over how many runs?
Group 05: Collaborative Dual-UAV Autonomous Surveillance System (MPSTME GUARDIAN)	2	2	2	2	2	1:45	How does your human-detection latency change with altitude or lighting, and how did you set the guard baseline?

SECTION 6 · Individual Learning Record & Research Consent *unmarked · one row per member · answer honestly*

These answers do not change your marks. They help improve the course and are used in anonymised research only if you consent in the sign-off below. Commit share: your commits ÷ the group's commits in your folder (git shortlog -sn -- <folder>).

Member	Hours I spent (total)	My declared role matched my actual work (1 = not at all, 5 = fully)	I learned a skill outside my role (robotics ↔ business)	Scripts I mainly wrote (file names)	My commit share (%)	Share of my writing first drafted with AI	Wrong AI facts / code I personally caught
Daneeka Abhijeet Roy (E057)	48	4	Yes	solar_defect_detector.py, defect_detection.py, solar_pv_roi.py, run_evaluation.py (AI-assisted; reviewed and run by me)	100	51–75 %	1–2

Member	Before this course I had used a physics simulator	The most valuable thing I learned (one line)	One thing I would change in this course (one line)
Daneeka Abhijeet Roy (E057)	Yes	A result in simulation only counts if you test it on cases you never tuned on, and say clearly what the model cannot tell you.	Give the business-model inputs and sources earlier, so the cost numbers can be checked sooner.

Track A: how this dossier becomes your 4-page IEEE paper

I. Introduction	Section 1: title, verified baseline friction, governing standards
II. Related work	Section 1.1: verified sources only (5+ with DOIs)
III. System model	Section 2: MJCF model, FSM, equations, control law
IV. Techno-economics & evaluation	Sections 3–4: CSBS model, FMEA, Monte Carlo results, statistics
V. Discussion & limitations	Sim-to-real gap, assumptions in the business model, AI audit lessons
Target venues (titles sheet)	A: agree a venue with Dr. Nanade · B: agree a venue with Dr. Nanade

Individual Academic Sign-off *every member answers and signs*

Answer each statement for yourself: (1) this work is original and every source and AI use is declared; (2) our model loads and our controller and business scripts run from the repository; (3) I understand our whole project and can explain my own part in the viva voce.

Member (Roll)	(1) Original work	(2) Code runs	(3) Viva-ready	Signature	Optional research consent*
Daneeka Abhijeet Roy (E057)	Yes	Yes	Yes	DANEEKA ROY	Yes

**Optional: may Dr. Nanade use anonymised data from this dossier and repository in education research publications? Your answer has no effect on your marks, and you can withdraw until results are published.*